Open Data Terms in Gravitational-Wave Physics: The LIGO and LVK Experience

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This one-pager is about how LIGO's release policy came about, how the LIGO-Virgo-KAGRA Collaboration (LVK) settled on terms, and what we've learned along the way. I had help from my colleague Jonah Kanner, who runs the LIGO open data program at Caltech.

First, a bit of background on who owns what. LIGO data belong to the LIGO Laboratory, which operates the detectors for the National Science Foundation (NSF). The LIGO Scientific Collaboration (LSC), more than a thousand people strong, analyzes the data and writes the papers. The LIGO Laboratory answers to the NSF for the data. Its Data Management Plan (https://dcc.ligo.org/LIGO-M1000066/public) dates back to around 2010, when the NSF started requiring these plans, and we've updated it for each observing run and data release. Starting in 2011, before building the open data service, the LSC surveyed the astronomy community and held a workshop to learn what scientists and other users outside our collaboration needed. Our first large data release came before any detection: in August 2014 we published the full dataset from Initial LIGO's S5 run, which contained no detected signals -- but it served as a testbed for our tools, documentation and user support a year before the first detection.

Our data come out on two tracks. Short stretches of data around significant events are released together with the paper that reports them. Bulk calibrated data from observing periods are released in blocks of up to a year, after a proprietary period. Proprietary periods in the past have been around 18 months, but the exact amount of time the data are held is evolving. We also release catalog products in Zenodo with Digital Object Identifiers (DOIs). The *Gravitational Wave Open Science Center* (GWOSC) makes the data available under CC BY 4.0. Nobody has to register or sign anything; we just ask users to acknowledge the project. For large-scale use, GWOSC data are also served through the Open Science Data Federation (OSDF), so people can run analyses on clusters worldwide without first downloading everything. LSC software generally goes out under the MIT License. Both licenses provide a lot of freedom to users, in the spirit of making our data and software easy to use and as open as possible.

The LVK never actually wrote a formal *joint* data policy, but in practice, LIGO, Virgo and KAGRA all follow the data release policy in the LIGO management plan. Each laboratory and collaboration answers to its own funders: the NSF for the LIGO Laboratory and the LSC; the French, Italian and Dutch agencies for EGO (the European Gravitational Observatory) and Virgo; and Japanese institutions for KAGRA. However, the partners do agree on the release dates, GWOSC as the repository, and the license and file formats. That would have been much harder without the data-sharing memorandum LIGO and Virgo signed in 2007, which KAGRA joined in 2019. Since that arrangement was already in place, we could treat open release as an extension of it.

The license itself has rarely been controversial. The harder part is getting the collaboration to agree on what data to release, and when. More data means more documentation and more community support, and not everyone agrees on how far to go. Each detector records one calibrated gravitational-wave (GW) data channel and roughly 200,000 diagnostic channels, and we release only a small subset of the diagnostic ones. People have also asked for uncalibrated GW data, but so far we haven't released it. We also hold back GW data from our engineering and commissioning periods, though there's sometimes debate about exactly when they end and observing periods begin.

The length of the proprietary period, where we hold back data for internal use only, is probably the touchiest question. Our main worry was that students and postdocs who had spent years on the instruments and analysis pipelines would get scooped if the data went public too soon after calibration, so we've adjusted the period from run to run. Outside groups did later find additional GW candidates in the O1 and O2 data, and we came to see that as good for the field.

In 2017, a group reported correlated noise in the GW150914 data implying a flaw in our published conclusions. Part of the problem was that the public material we had released didn't fully document how it had been processed, and the collaboration had to issue a formal response explaining why there was no issue with our published detections. The lesson the open data team took from this and similar episodes is that the public data files should be *exactly* the same ones we use internally. Every extra reprocessing step for release, whether it's removing bad times or converting formats, is another chance to introduce mistakes.

The use of two different licenses has created some smaller headaches. With MIT for software and CC BY for data, it isn't always obvious which one applies: a software tutorial hosted on the GWOSC webserver might be covered under either license, which could be confusing. On the other hand, the MIT license is really designed for software, and the CC licenses made more sense for instrument data products.

User support also took more effort than we expected. GWOSC users include high school students and hobbyists, and what they use most is the simplest material: how to open a file, how to make a plot. Someone must "own" answering the email, and there will always be requests for data in some other form. Having many examples and tutorials is essential, and we also hold open workshops. Some notes on this are at https://dcc.ligo.org/LIGO-G2201208/public

If I were advising a rare-event collaboration, I'd start with ownership: decide, in writing, who owns and curates the public data before the first release. Then write a short policy that ties releases to publications and to dates, so you're not renegotiating them every time. Use standard licenses as written and decide up front how you'll handle material that's part code and part data. Ask the DOE and the NSF early what each of them expects and put the cost of curation and support into your proposals. A small first release, built from the same files you actually use for analysis and with a few basic examples, is a good place to start.

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